

37. True or False

Statement:

If $F(x)$ is an antiderivative of $f(x)$, then

$$\int f(x) dx = F(x) + C$$

Answer:  **True**

Explanation

An **antiderivative** of $f(x)$ is a function $F(x)$ such that

$$F'(x) = f(x).$$

The **indefinite integral** represents the family of *all* antiderivatives of f . Since any two antiderivatives differ by a constant, we write

$$\int f(x) dx = F(x) + C,$$

where C is an arbitrary constant called the **constant of integration**.

Therefore, the statement is **True**.

38. True or False

Statement: If C denotes a constant of integration, the two formulas

$$\int \cos x dx = \sin x + C$$

and

$$\int \cos x dx = (\sin x + \pi) + C$$

are both correct equations.

Answer:  **True**

Explanation

We know

$$\frac{d}{dx}(\sin x) = \cos x.$$

Also,

$$\frac{d}{dx}(\sin x + \pi) = \cos x,$$

because the derivative of the constant π is zero.

Notice that

$$(\sin x + \pi) + C = \sin x + (C + \pi).$$

Since $C + \pi$ is still just another arbitrary constant, both expressions describe exactly the same family of antiderivatives.

Hence, the statement is **True**.

39. True or False

Statement: The function

$$f(x) = e^{-x} + 1$$

is a solution to the initial-value problem

$$\frac{dy}{dx} = -\frac{1}{e^x}, \quad y(0) = 1.$$

Answer: **✗ False**

Step 1: Check the differential equation

The given function is

$$y = e^{-x} + 1.$$

Differentiate:

$$y' = \frac{d}{dx}(e^{-x} + 1) = -e^{-x}.$$

Since

$$-\frac{1}{e^x} = -e^{-x},$$

the differential equation is satisfied.

Step 2: Check the initial condition

Evaluate at $x = 0$:

$$y(0) = e^0 + 1 = 1 + 1 = 2.$$

But the initial condition requires

$$y(0) = 1.$$

The condition fails.

Therefore, the statement is **False**.

40. True or False

Statement: Every integral curve of the slope field

$$\frac{dy}{dx} = \frac{1}{\sqrt{x^2 + 1}}$$

is the graph of an increasing function of x .

Answer:  **True**

Explanation

For every real number x ,

$$x^2 + 1 > 0.$$

Therefore,

$$\sqrt{x^2 + 1} > 0.$$

Hence,

$$\frac{dy}{dx} = \frac{1}{\sqrt{x^2 + 1}} > 0 \quad \text{for all } x.$$

A function whose derivative is positive everywhere is **increasing everywhere**.

Thus every integral curve of this slope field is increasing.

Therefore, the statement is **True**.

Final Answers

Problem	Answer
37	 True
38	 True
39	 False
40	 True

So the answers are:

37. *T*, 38. *T*, 39. *F*, 40. *T*

The idea is always the same:

1. Integrate the derivative to find the general solution.
2. Use the initial condition to determine the constant C .

43

(a)

$$\frac{dy}{dx} = \sqrt[3]{x}, \quad y(1) = 2$$

Since

$$\sqrt[3]{x} = x^{1/3},$$

integrate:

$$y = \int x^{1/3} dx = \frac{x^{4/3}}{4/3} + C = \frac{3}{4}x^{4/3} + C.$$

Use $y(1) = 2$:

$$2 = \frac{3}{4}(1)^{4/3} + C = \frac{3}{4} + C.$$

Thus,

$$C = 2 - \frac{3}{4} = \frac{5}{4}.$$

Therefore,

$$\boxed{y = \frac{3}{4}x^{4/3} + \frac{5}{4}}.$$

(b)

$$\frac{dy}{dt} = \sin t + 1, \quad y\left(\frac{\pi}{3}\right) = \frac{1}{2}$$

Integrate:

$$y = \int (\sin t + 1) dt = -\cos t + t + C.$$

Use the initial condition:

$$\frac{1}{2} = -\cos\left(\frac{\pi}{3}\right) + \frac{\pi}{3} + C.$$

Since

$$\cos\left(\frac{\pi}{3}\right) = \frac{1}{2},$$

we get

$$\frac{1}{2} = -\frac{1}{2} + \frac{\pi}{3} + C.$$

Hence,

$$C = 1 - \frac{\pi}{3}.$$

Therefore,

$$y = -\cos t + t + 1 - \frac{\pi}{3}.$$

(c)

$$\frac{dy}{dx} = \frac{x+1}{\sqrt{x}}, \quad y(1) = 0$$

Rewrite:

$$\frac{x+1}{\sqrt{x}} = x^{1/2} + x^{-1/2}.$$

Integrate:

$$y = \int (x^{1/2} + x^{-1/2}) dx = \frac{2}{3}x^{3/2} + 2x^{1/2} + C.$$

Use $y(1) = 0$:

$$0 = \frac{2}{3} + 2 + C.$$

Thus,

$$C = -\frac{8}{3}.$$

Therefore,

$$y = \frac{2}{3}x^{3/2} + 2\sqrt{x} - \frac{8}{3}.$$

44

(a)

$$\frac{dy}{dx} = \frac{1}{(2x)^3}, \quad y(1) = 0$$

Since

$$\frac{1}{(2x)^3} = \frac{1}{8x^3} = \frac{1}{8}x^{-3},$$

integrate:

$$y = \frac{1}{8} \int x^{-3} dx = \frac{1}{8} \left(\frac{x^{-2}}{-2} \right) + C = -\frac{1}{16x^2} + C.$$

Use $y(1) = 0$:

$$0 = -\frac{1}{16} + C.$$

So

$$C = \frac{1}{16}.$$

Hence,

$$\boxed{y = \frac{1}{16} - \frac{1}{16x^2}}.$$

(b)

$$\frac{dy}{dt} = \sec^2 t - \sin t, \quad y\left(\frac{\pi}{4}\right) = 1$$

Integrate:

$$y = \tan t + \cos t + C.$$

Use the condition:

$$1 = \tan \frac{\pi}{4} + \cos \frac{\pi}{4} + C.$$

Since

$$\tan \frac{\pi}{4} = 1, \quad \cos \frac{\pi}{4} = \frac{\sqrt{2}}{2},$$

we obtain

$$1 = 1 + \frac{\sqrt{2}}{2} + C.$$

Thus,

$$C = -\frac{\sqrt{2}}{2}.$$

Therefore,

$$\boxed{y = \tan t + \cos t - \frac{\sqrt{2}}{2}}.$$

(c)

$$\frac{dy}{dx} = x^2 \sqrt{x^3}, \quad y(0) = 0$$

Since

$$\sqrt{x^3} = x^{3/2},$$

we have

$$\frac{dy}{dx} = x^{7/2}.$$

Integrate:

$$y = \int x^{7/2} dx = \frac{x^{9/2}}{9/2} + C = \frac{2}{9} x^{9/2} + C.$$

Use $y(0) = 0$:

$$0 = 0 + C,$$

so

$$C = 0.$$

Thus,

$$\boxed{y = \frac{2}{9} x^{9/2}}.$$

45

(a)

$$\frac{dy}{dx} = 4e^x, \quad y(0) = 1$$

Integrate:

$$y = 4e^x + C.$$

Use $y(0) = 1$:

$$1 = 4 + C.$$

Hence,

$$C = -3.$$

Therefore,

$$\boxed{y = 4e^x - 3}.$$

(b)

$$\frac{dy}{dt} = \frac{1}{t}, \quad y(-1) = 5$$

Integrate:

$$y = \ln |t| + C.$$

Use $t = -1$:

$$5 = \ln |-1| + C = \ln 1 + C = 0 + C.$$

Thus,

$$C = 5.$$

Hence,

$$\boxed{y = \ln |t| + 5}.$$

46

(a)

$$\frac{dy}{dt} = \frac{3}{\sqrt{1-t^2}}, \quad y\left(\frac{\sqrt{3}}{2}\right) = 0$$

Recall:

$$\int \frac{1}{\sqrt{1-t^2}} dt = \arcsin t + C.$$

Therefore,

$$y = 3 \arcsin t + C.$$

Use the condition:

$$0 = 3 \arcsin\left(\frac{\sqrt{3}}{2}\right) + C.$$

Since

$$\arcsin\left(\frac{\sqrt{3}}{2}\right) = \frac{\pi}{3},$$

we get

$$C = -\pi.$$

Thus,

$$\boxed{y = 3 \arcsin t - \pi}.$$

(b)

$$\frac{dy}{dx} = \frac{x^2 - 1}{x^2 + 1}, \quad y(1) = \frac{\pi}{2}$$

Rewrite:

$$\frac{x^2 - 1}{x^2 + 1} = 1 - \frac{2}{x^2 + 1}.$$

Integrate:

$$y = x - 2 \arctan x + C.$$

Use $y(1) = \frac{\pi}{2}$:

$$\frac{\pi}{2} = 1 - 2 \left(\frac{\pi}{4} \right) + C = 1 - \frac{\pi}{2} + C.$$

Hence,

$$C = \pi - 1.$$

Therefore,

$$\boxed{y = x - 2 \arctan x + \pi - 1}.$$

Final Answers

$$\begin{aligned} 43(a) \quad y &= \frac{3}{4}x^{4/3} + \frac{5}{4} \\ 43(b) \quad y &= -\cos t + t + 1 - \frac{\pi}{3} \\ 43(c) \quad y &= \frac{2}{3}x^{3/2} + 2\sqrt{x} - \frac{8}{3} \\ 44(a) \quad y &= \frac{1}{16} - \frac{1}{16x^2} \\ 44(b) \quad y &= \tan t + \cos t - \frac{\sqrt{2}}{2} \\ 44(c) \quad y &= \frac{2}{9}x^{9/2} \\ 45(a) \quad y &= 4e^x - 3 \\ 45(b) \quad y &= \ln |t| + 5 \\ 46(a) \quad y &= 3 \arcsin t - \pi \\ 46(b) \quad y &= x - 2 \arctan x + \pi - 1 \end{aligned}$$

For Problems **47–50**, remember:

- Velocity is the derivative of position:

$$v(t) = s'(t).$$

- To find the position function $s(t)$, integrate the velocity:

$$s(t) = \int v(t) dt + C.$$

- Then use the given initial condition to find C .

47.

Given

$$v(t) = 32t, \quad s(0) = 20.$$

Since

$$s'(t) = 32t,$$

integrate:

$$s(t) = \int 32t dt = 16t^2 + C.$$

Use $s(0) = 20$:

$$20 = 16(0)^2 + C,$$

so

$$C = 20.$$

Therefore,

$s(t) = 16t^2 + 20.$

48.

Given

$$v(t) = \cos t, \quad s(0) = 2.$$

Integrate:

$$s(t) = \int \cos t dt = \sin t + C.$$

Use $s(0) = 2$:

$$2 = \sin 0 + C = 0 + C.$$

Thus,

$$C = 2.$$

Therefore,

$$\boxed{s(t) = \sin t + 2.}$$

49.

Given

$$v(t) = 3\sqrt{t}, \quad s(4) = 1.$$

Rewrite:

$$3\sqrt{t} = 3t^{1/2}.$$

Integrate:

$$s(t) = 3 \int t^{1/2} dt = 3 \left(\frac{t^{3/2}}{3/2} \right) + C = 2t^{3/2} + C.$$

Use $s(4) = 1$:

Since

$$4^{3/2} = (\sqrt{4})^3 = 2^3 = 8,$$

we get

$$1 = 2(8) + C = 16 + C.$$

Hence,

$$C = -15.$$

Therefore,

$$\boxed{s(t) = 2t^{3/2} - 15.}$$

50.

Given

$$v(t) = 3e^t, \quad s(1) = 0.$$

Integrate:

$$s(t) = \int 3e^t dt = 3e^t + C.$$

Use $s(1) = 0$:

$$0 = 3e + C.$$

Therefore,

$$C = -3e.$$

Thus,

$$\boxed{s(t) = 3e^t - 3e.}$$

51.

Find a general form of a function whose second derivative is

$$\sqrt{x}.$$

That is,

$$f''(x) = \sqrt{x} = x^{1/2}.$$

Integrate once:

$$f'(x) = \int x^{1/2} dx = \frac{x^{3/2}}{3/2} + C_1 = \frac{2}{3}x^{3/2} + C_1.$$

Integrate again:

$$f(x) = \int \left(\frac{2}{3}x^{3/2} + C_1 \right) dx.$$

Therefore,

$$f(x) = \frac{2}{3} \cdot \frac{x^{5/2}}{5/2} + C_1x + C_2.$$

Simplifying,

$$\frac{2}{3} \cdot \frac{2}{5} = \frac{4}{15},$$

so

$$\boxed{f(x) = \frac{4}{15}x^{5/2} + C_1x + C_2.}$$

52.

Find a function f such that

$$f''(x) = x + \cos x,$$

with

$$f(0) = 1, \quad f'(0) = 2.$$

Step 1: Find $f'(x)$

Integrate:

$$f'(x) = \int (x + \cos x) dx = \frac{x^2}{2} + \sin x + C_1.$$

Use $f'(0) = 2$:

$$2 = 0 + 0 + C_1,$$

so

$$C_1 = 2.$$

Hence,

$$f'(x) = \frac{x^2}{2} + \sin x + 2.$$

Step 2: Find $f(x)$

Integrate again:

$$f(x) = \int \left(\frac{x^2}{2} + \sin x + 2 \right) dx.$$

Thus,

$$f(x) = \frac{x^3}{6} - \cos x + 2x + C_2.$$

Use $f(0) = 1$:

$$1 = 0 - \cos 0 + 0 + C_2.$$

Since $\cos 0 = 1$,

$$1 = -1 + C_2,$$

which gives

$$C_2 = 2.$$

Therefore,

$$f(x) = \frac{x^3}{6} - \cos x + 2x + 2.$$

Final Answers

$$\begin{aligned} 47. s(t) &= 16t^2 + 20, \\ 48. s(t) &= \sin t + 2, \\ 49. s(t) &= 2t^{3/2} - 15, \\ 50. s(t) &= 3e^t - 3e, \\ 51. f(x) &= \frac{4}{15}x^{5/2} + C_1x + C_2, \\ 52. f(x) &= \frac{x^3}{6} - \cos x + 2x + 2. \end{aligned}$$

The phrase "the slope of the curve" means the derivative

$$\frac{dy}{dx}.$$

So our strategy is:

1. Write the differential equation.
2. Integrate to get the general equation of the curve.
3. Use the given point or condition to find the constant.

53.

At each point (x, y) on the curve, the slope is

$$2x + 1,$$

and the curve passes through $(-3, 0)$.

Step 1: Write the differential equation

$$\frac{dy}{dx} = 2x + 1.$$

Step 2: Integrate

$$y = \int (2x + 1) dx = x^2 + x + C.$$

Step 3: Use the point $(-3, 0)$

Substitute $x = -3$, $y = 0$:

$$0 = (-3)^2 + (-3) + C.$$

Thus,

$$0 = 9 - 3 + C = 6 + C,$$

so

$$C = -6.$$

Answer

$y = x^2 + x - 6.$

54.

At each point, the slope is

$$(x + 1)^2,$$

and the curve passes through $(-2, 8)$.

Step 1

$$\frac{dy}{dx} = (x + 1)^2.$$

Expand:

$$(x + 1)^2 = x^2 + 2x + 1.$$

Step 2: Integrate

$$y = \int (x^2 + 2x + 1) dx = \frac{x^3}{3} + x^2 + x + C.$$

Step 3: Use $(-2, 8)$

$$8 = \frac{(-2)^3}{3} + (-2)^2 + (-2) + C.$$

That is,

$$8 = -\frac{8}{3} + 4 - 2 + C.$$

Since

$$4 - 2 = 2 = \frac{6}{3},$$

we get

$$8 = -\frac{2}{3} + C.$$

Therefore,

$$C = 8 + \frac{2}{3} = \frac{26}{3}.$$

Answer

$$y = \frac{x^3}{3} + x^2 + x + \frac{26}{3}.$$

55.

At each point, the slope is

$$-\sin x,$$

and the curve passes through $(0, 2)$.

Step 1

$$\frac{dy}{dx} = -\sin x.$$

Step 2: Integrate

Since

$$\frac{d}{dx}(\cos x) = -\sin x,$$

we have

$$y = \cos x + C.$$

Step 3: Use $(0, 2)$

$$2 = \cos 0 + C = 1 + C.$$

Thus,

$$C = 1.$$

Answer

$$y = \cos x + 1.$$

56.

At each point, the slope equals the square of the distance between the point and the y -axis; the point $(-1, 2)$ is on the curve.

Step 1: Interpret the condition

The distance from a point (x, y) to the y -axis is

$$|x|.$$

Its square is

$$x^2.$$

Therefore,

$$\frac{dy}{dx} = x^2.$$

Step 2: Integrate

$$y = \int x^2 dx = \frac{x^3}{3} + C.$$

Step 3: Use $(-1, 2)$

$$2 = \frac{(-1)^3}{3} + C = -\frac{1}{3} + C.$$

Hence,

$$C = 2 + \frac{1}{3} = \frac{7}{3}.$$

Answer

$$\boxed{y = \frac{x^3}{3} + \frac{7}{3}}.$$

57.

At each point on the curve,

$$\frac{d^2y}{dx^2} = 6x,$$

and

$$y = 5 - 3x$$

is tangent to the curve at the point where

$$x = 1.$$

The tangent line provides two pieces of information:

- The point on the curve when $x = 1$,
 - The slope at that point.
-

Step 1: Find y'

Integrate:

$$y' = \int 6x \, dx = 3x^2 + C_1.$$

Step 2: Use the tangent slope

The tangent line is

$$y = 5 - 3x.$$

Its slope is

$$-3.$$

Therefore,

$$y'(1) = -3.$$

Substitute into y' :

$$-3 = 3(1)^2 + C_1.$$

Thus,

$$C_1 = -6.$$

Hence,

$$y' = 3x^2 - 6.$$

Step 3: Find y

Integrate again:

$$y = \int (3x^2 - 6) \, dx = x^3 - 6x + C_2.$$

Step 4: Use the point from the tangent line

Since the curve and tangent line touch at $x = 1$,

$$y(1) = 5 - 3(1) = 2.$$

Substitute:

$$2 = 1 - 6 + C_2.$$

Therefore,

$$C_2 = 7.$$

Answer

$$y = x^3 - 6x + 7.$$

Final Answers

$$53. y = x^2 + x - 6,$$

$$54. y = \frac{x^3}{3} + x^2 + x + \frac{26}{3},$$

$$55. y = \cos x + 1,$$

$$56. y = \frac{x^3}{3} + \frac{7}{3},$$

$$57. y = x^3 - 6x + 7.$$

These problems all use the same powerful idea: a derivative tells you how a function changes, and integration lets you reconstruct the original curve from that information.